

HDOC Day-29

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge 2 arrays.
4. Exit

● C Program:

```
#include <stdio.h>

int main()
{
    int choice;

    int a[100], b[100], merged[200];

    int n, m, i, key, found;

    int even, odd, positive, negative, zero;

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Count Even, Odd, Positive, Negative and Zero\n");
        printf("2. Linear Search in an Array\n");
        printf("3. Merge Two Arrays\n");
        printf("4. Exit\n");
        printf("===== \n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
```



```
}  
}
```

```
printf("\nEven numbers = %d\n", even);  
printf("Odd numbers = %d\n", odd);  
printf("Positive numbers = %d\n", positive);  
printf("Negative numbers = %d\n", negative);  
printf("Zero elements = %d\n", zero);
```

```
break;
```

case 2:

```
printf("\nEnter the number of elements: ");  
scanf("%d", &n);
```

```
printf("Enter %d elements:\n", n);
```

```
for (i = 0; i < n; i++)  
    scanf("%d", &a[i]);
```

```
printf("Enter the element to search: ");  
scanf("%d", &key);
```

```
found = 0;
```

```
for (i = 0; i < n; i++)  
{  
    if (a[i] == key)
```

```
{  
    printf("Element found at position %d.\n", i + 1);  
    found = 1;  
    break;  
}  
}
```

```
if (found == 0)  
    printf("Element not found in the array.\n");
```

```
break;
```

case 3:

```
printf("\nEnter the size of first array: ");  
scanf("%d", &n);
```

```
printf("Enter %d elements:\n", n);
```

```
for (i = 0; i < n; i++)  
    scanf("%d", &a[i]);
```

```
printf("Enter the size of second array: ");  
scanf("%d", &m);
```

```
printf("Enter %d elements:\n", m);  
for (i = 0; i < m; i++)  
    scanf("%d", &b[i]);
```

```
/* Copy first array */
```

```
    for (i = 0; i < n; i++)
        merged[i] = a[i];

    /* Copy second array */
    for (i = 0; i < m; i++)
        merged[n + i] = b[i];

    printf("\nMerged array:\n");

    for (i = 0; i < n + m; i++)
        printf("%d ", merged[i]);

    printf("\n");

    break;

case 4:
    printf("\nProgram exited successfully.\n");
    break;

default:
    printf("\nInvalid choice! Please enter 1 to 4.\n");
}

} while (choice != 4);

return 0;
}
```

Test Case Table:

Test Case	Choice	Input	Expected Output
1	1	10 -5 0 7 -2 8	E= 3, O= 2, P= 3, N= 2, Z= 1
2	2	10 20 30 40 50	Element found at position 3
3	3	First array elements= 1 2 3, Second array elements= 4 5 6 7	Merged array= 1234567
4	4	Program exits successfully	Program exited successfully
5	5	-	Invalid choice

Compilation and Execution:

Test Case 1:

```
(kali㉿kali)-[~]
$ ./a.out

_____ MENU _____
1. Count Even, Odd, Positive, Negative and Zero
2. Linear Search in an Array
3. Merge Two Arrays
4. Exit
_____

Enter your choice: 1

Enter the number of elements: 6
Enter 6 elements:
10 -5 0 7 -2 8

Even numbers = 3
Odd numbers = 2
Positive numbers = 3
Negative numbers = 2
Zero elements = 1
```

Test Case 2:

```
———— MENU ————
1. Count Even, Odd, Positive, Negative
2. Linear Search in an Array
3. Merge Two Arrays
4. Exit

Enter your choice: 2

Enter the numbers of elements: 5
Enter 5 elements:
10 20 30 40 50
Enter the element to search:30
Element found at position 3.
```

Test Case 3:

```
———— MENU ————
1. Count Even, Odd, Positive, Negative and Zero
2. Linear Search in an Array
3. Merge Two Arrays
4. Exit

Enter your choice: 3

Enter the size of first array: 3
Enter 3 elemnts:
1 2 3
Enter the size of second array: 4
Enter 4 elements:
4 5 6 7

Merged array:
1234567
```

Test Case 4:

```
———— MENU ————
1. Count Even, Odd, Positive, Negative and Zero
2. Linear Search in an Array
3. Merge Two Arrays
4. Exit

Enter your choice: 4

Program exited successfully.
```

Test Case 5:

```
(kali㉿kali)-[~]  
$ ./a.out
```

_____ MENU _____

1. Count Even, Odd, Positive, Negative and Zero
2. Linear Search in an Array
3. Merge Two Arrays
4. Exit

Enter your choice: 5

Invalid choice! Please enter 1 to 4.